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A CASE STUDY REPORT

ON

GLOBAL AND LOCAL PAYMENT SYSTEMS

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A CASE STUDY REPORT SUBMITTED TO THE DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY IN PARTIAL FULFILMENT OF THE REQUIREMENTS FOR THE DEGREE OF BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

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EXECUTIVE SUMMARY

Payment systems constitute the operational and technical backbone through which monetary value moves between individuals, businesses, financial institutions, and governments. This report presents a comprehensive, technically grounded case study of two broad categories of payment infrastructure: global payment systems, which facilitate messaging and settlement across national borders, and local (domestic) payment systems, which are designed, regulated, and operated within a single jurisdiction to serve retail, government, and interbank transactions. The report examines the architectural distinctions between messaging networks and settlement mechanisms, the four party and three party card models, real time gross settlement (RTGS) systems, and retail instant payment rails, before applying this framework to a detailed technical review of the major global networks, SWIFT, Visa, Mastercard, PayPal, Stripe, Alipay, and WeChat Pay, and a set of local case studies covering India's Unified Payments Interface (UPI), Brazil's Pix, and Nepal's domestic payment ecosystem centred on Nepal Rastra Bank, Nepal Clearing House Limited (NCHL), connectIPS, eSewa, and Khalti.

The findings indicate a persistent and structurally explainable divergence between the two categories. Global systems continue to dominate cross border trade, remittances, and multi-currency commerce because they alone provide the interoperability, currency conversion, and multi-jurisdictional regulatory coverage that cross border transactions require; however, they remain comparatively costly and, in many corridors, comparatively slow, with settlement measured in hours or days rather than seconds. By contrast, local instant payment systems, have achieved extraordinary adoption in emerging economies precisely because they are optimised for a single regulatory environment, a single currency, and a mobile first user base, allowing near zero cost, twenty four hour, seven day settlement that global rails cannot presently match domestically. Nepal exhibits this same pattern: connectIPS, NCHL-IPS, and the NEPALPAY QR standard now provide a genuinely interoperable, low cost domestic settlement layer, while SWIFT and the international card networks remain the effective monopoly channel for the country's remittance inflows and cross border merchant settlement.

The report further finds that interoperability, rather than substitution, is the emerging strategic frontier. Bilateral and multilateral linkages between domestic fast payment systems, exemplified by early India to Singapore and India to United Arab Emirates UPI linkages, alongside the modernisation of SWIFT through the gpi initiative and the industry wide migration to the ISO 20022 messaging standard, suggest that the future architecture of global

commerce will combine locally optimised retail rails with a modernised global messaging and settlement layer rather than a wholesale replacement of one by the other. For Nepal specifically, the report recommends continued investment in NCHL's interoperable infrastructure, deeper integration between connectIPS and international remittance corridors, strengthened fraud and cybersecurity controls commensurate with the irreversibility of instant transfers, and sustained regulatory attention from Nepal Rastra Bank to licensing, interoperability, and consumer protection as the sector continues to expand.

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CHAPTER 1: INTRODUCTION

A payment system is a set of instruments, procedures, rules, and institutional arrangements that facilitate the transfer of monetary value between parties. At a technical level, every payment involves two logically distinct processes: messaging, the exchange of instructions specifying who is paying whom, how much, and in what currency; and settlement, the actual movement of value between the accounts of the paying and receiving institutions, typically at a central bank or a designated settlement agent. Many payment systems that appear to the end user as a single seamless transaction are, beneath the surface, a coordinated sequence of messaging and settlement steps carried out by several distinct institutions operating under a common technical and legal framework.

Payment systems can broadly be classified into two categories. Global payment systems are networks, messaging standards, and settlement arrangements that operate across national borders, connecting banks and financial institutions worldwide and accommodating multiple currencies and regulatory regimes. Local payment systems, by contrast, are designed, licensed, and operated within a single country to serve domestic retail, government, and interbank transactions, typically under the direct oversight of a national central bank. Over the past two decades, digitalisation, smartphone penetration, and central bank policy initiatives have significantly reshaped how individuals and businesses transact, with emerging economies in particular developing domestic real time payment rails that now handle a substantial share of daily transactions, in some cases surpassing traditional card networks in transaction volume.

1.1 Architectural and Functional Foundations

At the architectural level, a distinction must be drawn between three related but separable functions: messaging, clearing, and settlement. Messaging systems, of which SWIFT is the paradigmatic example, transmit standardised instructions between financial institutions but do not themselves move funds. Clearing is the process by which the mutual obligations arising from a batch of transactions are calculated and netted, often by a clearing house acting as an intermediary between many participating banks. Settlement is the final, irrevocable transfer of value that discharges the obligations identified during clearing, and it typically occurs across accounts that participating institutions hold at a central bank.

A further foundational distinction separates real time gross settlement (RTGS) systems, in which each transaction is settled individually and immediately, from deferred net settlement

(DNS) systems, in which transactions are accumulated over a period and settled in net batches at fixed intervals. RTGS systems, typically operated directly by a central bank for large value interbank transfers, eliminate settlement risk between the moment of transaction and final settlement but require each participating bank to maintain sufficient liquidity to cover gross outflows throughout the day. DNS systems economise on liquidity by netting offsetting flows before settlement but introduce a window of settlement risk during which a participating institution could, in principle, fail before its net obligations are discharged.

Retail instant payment systems, such as India's UPI, Brazil's Pix, and Nepal's NCHL-IPS, occupy a distinct architectural category: they are designed for high volume, low value, person to person and person to merchant transactions, and they achieve settlement finality within seconds by combining real time messaging with pre funded or centrally guaranteed settlement arrangements that do not require each individual transaction to pass through the full RTGS liquidity management process used for wholesale interbank transfers. Card networks, in turn, separate authorisation, an near instant check that a transaction is permitted and funds or credit are available, from clearing and settlement, which for many card transactions still occurs in daily or multi day batches even though the consumer experiences authorisation as instantaneous.

1.2 Scope of the Report

The scope of this report is limited to retail and interbank electronic payment systems. It does not extend to physical cash handling, informal or unregulated remittance channels, or specialised wholesale settlement systems used exclusively for large value interbank transfers, except where relevant to explain the broader payment infrastructure of a country. Geographically, the report examines major global networks as a point of technical reference before narrowing to detailed case studies of India, Brazil, and, in particular, Nepal, whose domestic payment ecosystem forms the primary local focus of the analysis. The report does not attempt a real time market data census of transaction volumes or fees; rather, it draws on central bank disclosures, payment system operator publications, and industry reporting to describe general, structurally grounded patterns that a student, practitioner, or policymaker can reasonably rely upon.

1.3 Methodology

This report is a desk based case study, compiled from publicly available information, including central bank publications and directives, payment system operator disclosures, SWIFT and card network documentation, Bank for International Settlements (BIS) reports, industry benchmarks, and academic and trade press coverage of the payments sector. Comparative tables are used throughout to summarise the characteristics of each system on a consistent basis across five evaluation criteria: cost, settlement speed, accessibility and financial inclusion impact, regulatory oversight, and interoperability. Because payment system fee schedules, transaction volumes, and technical specifications evolve continuously, and because central banks and network operators update their disclosures on differing schedules, figures relating to cost, volume, and coverage in this report should be treated as indicative of the general pattern observed at the time of writing rather than as real time data, and readers requiring current figures are encouraged to consult the originating central bank, network operator, or standards body directly.

1.4 Research Questions and Problem Definition

The report is structured around the following guiding research questions:

- What are the principal architectural and regulatory categories into which payment systems can be classified, and what trade offs does each category impose in terms of cost, speed, and control?
- How do the major global messaging and card networks function technically, and what structural factors explain their continued dominance in cross border commerce?
- How have domestic instant payment systems in India, Brazil, and Nepal been designed, and what factors explain their rapid adoption relative to global alternatives for retail transactions?
- How do global and local payment systems compare systematically on the basis of cost, settlement speed, accessibility, regulatory oversight, and interoperability?
- What security, fraud, and operational risks are distinctive to each category of system, and what mitigation strategies are being applied by operators and regulators?

- What emerging trends, including cross border fast payment linkages, central bank digital currencies, and ISO 20022 migration, are likely to shape the future relationship between global and local payment infrastructure?

The underlying problem this report addresses is one of interoperability barriers, foreign exchange cost, digital inclusion, and security: organisations and individuals frequently interact with payment infrastructure without a structured understanding of why a domestic QR payment settles in seconds at negligible cost while a cross border remittance can take days and carry a fee exceeding five percent of the transferred value. By presenting a systematic classification, detailed technical case studies, and a comparative framework grounded in the five evaluation criteria above, this report aims to equip students, practitioners, and policymakers with a clearer basis for understanding, and where relevant improving, this divide.

The remainder of this report proceeds as follows. Chapter 2 establishes the architectural and categorical classification used throughout the analysis. Chapter 3 sets out the study's objectives in itemised form. Chapter 4 examines the major global payment systems in technical depth. Chapter 5 narrows the focus to a sector by sector case study of local systems, with particular attention to Nepal. Chapter 6 presents a structured comparative analysis across five evaluation criteria. Chapter 7 discusses risks, security, and operational challenges facing both categories of system. Chapter 8 looks ahead to emerging trends, and the report closes with a strategic conclusion and stakeholder specific recommendations in Chapters 9 and 10.

CHAPTER 2: ARCHITECTURAL AND CATEGORICAL CLASSIFICATION

Before examining individual systems, it is useful to classify payment infrastructure along several independent dimensions. This classification provides the analytical framework applied throughout the comparative analysis presented later in this report, and it clarifies that the categories described are not mutually exclusive: a single country's payment ecosystem typically combines several of these models simultaneously, with global card networks operating alongside domestic real time transfer systems and privately operated digital wallets.

Table 1: Classification Framework of Payment Systems

Classification Basis	Categories	Representative Examples
Geographic scope	Global (cross border) / Local (domestic)	SWIFT, Visa (global); UPI, connectIPS (local)
Settlement speed	Real time gross settlement / Deferred net settlement / Retail instant	RTGS systems (gross); card network batches (deferred net); Pix, NCHL-IPS (instant)
Underlying mechanism	Card based / Account to account (A2A) / Wallet based	Visa, Mastercard (card); UPI, SEPA (A2A); eSewa, Khalti (wallet)
Operating authority	Central bank operated / Privately operated / Public private	FedNow (central bank); PayPal (private); connectIPS (public private, NCHL)

2.1 Card Networks versus Account to Account (A2A) Rails

Card networks operate on a four party model, comprising the cardholder, the issuing bank, the merchant, and the acquiring bank, connected by a central switch operated by the network itself, such as Visa or Mastercard, or a three party model, such as American Express, in which the network acts as both issuer and acquirer. Every card transaction passes through discrete authorisation, clearing, and settlement stages, and interchange, the fee paid by the acquiring bank to the issuing bank on each transaction, is centrally set by the network within regulatory limits that vary by jurisdiction and transaction type.

Account to account (A2A) rails, by contrast, move value directly between the payer's and payee's bank accounts without an intermediary card product, typically initiated through a mobile number, a virtual payment address, or a QR code. Because A2A systems such as UPI and connectIPS bypass the card interchange model entirely, they can operate at substantially lower cost, and in several cases at zero cost to the end user, particularly where the operating

authority is a central bank or a public private utility with a financial inclusion mandate rather than a profit maximising commercial network.

2.2 Centralized Settlement versus Distributed or Decentralized Models

The overwhelming majority of both global and local payment systems in commercial operation today rely on centralised settlement, in which a single trusted intermediary, typically a central bank, a clearing house, or a network operator, maintains the authoritative ledger of account balances and executes settlement on that basis. Centralised settlement offers strong finality guarantees, straightforward dispute resolution, and clear regulatory accountability, at the cost of concentrating operational and systemic risk in a small number of critical infrastructure providers.

Distributed or decentralised models, most closely associated with blockchain based cryptocurrency networks, distribute the ledger and the validation of transactions across many independent participants rather than a single central authority. While such models remain outside the mainstream retail and interbank payment infrastructure examined in this report, central banks in numerous jurisdictions are exploring hybrid designs, discussed further in Chapter 8, that borrow selected distributed ledger techniques, such as programmable settlement and atomic delivery versus payment, while retaining central bank oversight of the underlying unit of account.

2.3 Regulatory and Licensing Tiers

Most jurisdictions distinguish, at a regulatory level, between payment system operators (PSOs), entities licensed to operate the underlying clearing and settlement infrastructure itself, and payment service providers (PSPs), entities licensed to offer payment services, such as digital wallets or merchant acquiring, to end users on top of that infrastructure. In Nepal, this distinction is codified in the Payment and Settlement Act, 2019, and the associated Nepal Rastra Bank bylaws, under which Nepal Clearing House Limited is licensed as a payment system operator responsible for NCHL-IPS and connectIPS, while eSewa and Khalti are licensed as payment service providers that connect to this underlying infrastructure to offer wallet based services to consumers and merchants. This two tier licensing structure allows a central bank to concentrate prudential and systemic risk oversight on a small number of infrastructure operators while permitting a competitive, innovation driven market of consumer facing service providers to operate on top of a common, regulated rail, a structure mirrored in

India's separation between NPCI, the payment system operator for UPI, and the many bank and non bank applications that connect to it.

CHAPTER 3: OBJECTIVES OF THE STUDY

The primary objectives of this study are:

- To examine the structure, technical architecture, and comparative characteristics of major global payment systems, including SWIFT, the international card networks, and global digital payment platforms.
- To examine the structure, technical architecture, and adoption drivers of major local and regional payment systems, with detailed reference to India's UPI, Brazil's Pix, and Nepal's domestic payment infrastructure.
- To compare global and local payment systems systematically on the basis of total transaction cost, settlement speed and availability, accessibility and financial inclusion impact, regulatory oversight and cross border sovereignty, and interoperability.
- To identify the security, fraud, and operational risks facing both global and local payment infrastructure, and to propose practical mitigation strategies for each.
- To assess emerging global and Nepal specific trends, including cross border fast payment linkages, central bank digital currencies, ISO 20022 migration, and artificial intelligence driven fraud mitigation, and their likely influence on future payment system design.

The secondary objectives of this study are:

- To document Nepal's regulatory and institutional framework for payment systems, centred on Nepal Rastra Bank, the Payment and Settlement Act, and Nepal Clearing House Limited.
- To catalogue the licensed payment service providers and operators active in Nepal and their respective technology stacks and key features.
- To provide differentiated, actionable recommendations for four distinct stakeholder groups: central banks and regulators, domestic payment operators, commercial banks and financial institutions, and merchants and end users.

CHAPTER 4: GLOBAL PAYMENT SYSTEMS AND INFRASTRUCTURE ANALYSIS

Global payment systems are the networks and messaging standards that enable financial institutions in different countries to exchange payment instructions and settle transactions across currencies and regulatory jurisdictions. They are essential for international trade, remittances, foreign travel, and cross border electronic commerce, and typically involve multiple intermediary institutions, correspondent banks, card acquirers, and foreign exchange counterparties, before funds reach the final beneficiary. This chapter examines the major global systems in technical depth before presenting comparative summary and benchmark tables.

4.1 SWIFT: Messaging Architecture, ISO 20022, and Correspondent Banking Friction

SWIFT (Society for Worldwide Interbank Financial Telecommunication) is a global messaging network used by banks and financial institutions to securely transmit standardised instructions related to financial transactions. Founded in 1973 and headquartered in Belgium, SWIFT connects thousands of financial institutions across more than two hundred countries and territories. Critically, SWIFT itself does not hold funds or settle payments; it provides the standardised messaging infrastructure, historically based on the MT (Message Type) standard, through which banks instruct one another to move money, with actual settlement occurring through correspondent banking relationships, in which one bank holds an account (a nostro or vostro account) with another, or through national RTGS systems for domestic legs of a cross border transaction.

The industry is currently migrating from the legacy MT message format to ISO 20022, a richer, XML based messaging standard that carries substantially more structured data per transaction, including detailed remittance information, ultimate originator and beneficiary identifiers, and standardised purpose codes. This migration, coordinated across SWIFT and numerous domestic RTGS operators on a phased timetable, is intended to improve straight through processing rates, reduce the manual investigation of failed or flagged payments, and provide a common data language across previously incompatible national systems.

A cross border payment routed through correspondent banking frequently passes through two, three, or more intermediary banks before reaching its destination, each of which may deduct a handling fee and apply its own compliance screening, a structure that historically produced unpredictable total cost, unpredictable delivery time, and limited visibility for the originating

customer as to where a payment stood in the chain at any given moment. SWIFT gpi (Global Payments Innovation), introduced to address these frictions, requires participating banks to attach a unique end to end transaction reference to each payment and to confirm credit to the beneficiary account, allowing the originating bank and, in many cases, the originating customer, to track a payment's status across the correspondent chain in a manner broadly analogous to parcel tracking, while committing participating banks to same day use of funds within their own processing window.

4.2 International Card Schemes: The Four Party Model and Cross Border Friction

Visa and Mastercard operate global card networks that allow consumers to make payments using debit or credit cards across borders, with standardised technical rules for authorisation, clearing, settlement, and dispute resolution administered centrally by the network while card issuance and merchant acquiring are carried out by thousands of independent participating banks. A card issued in one country can typically be used to make a purchase in another country through the same network, offering merchants and consumers a broadly consistent experience regardless of location, a property that has made these networks the default rail for international card not present and point of sale commerce.

This convenience carries a measurable cost. Cross border card transactions typically attract a network assessment fee of approximately one percent of transaction value in addition to a merchant service fee of one to three percent charged by the acquiring bank, and a further currency conversion margin where the transaction currency differs from the cardholder's billing currency, contributing to profit margins that are among the highest of any publicly listed industry sector. Both companies operate on a four party model, Visa and Mastercard, that separates card issuance from merchant acquiring and interchange setting, or a three party model, American Express, in which the network performs both roles directly, allowing thousands of banks worldwide to participate in a single technically and commercially standardised network

4.3 Global Digital Platforms

Beyond bank led messaging and card networks, global technology platforms such as PayPal and Stripe facilitate cross border consumer and business payments, typically layered on top of existing card and bank rails rather than replacing them; PayPal provides a consumer facing

digital wallet and merchant checkout product, while Stripe is positioned primarily as developer facing payment infrastructure that merchants and platforms integrate directly into their own checkout flows through an application programming interface, abstracting away much of the underlying card acquiring and compliance complexity.

Alongside these Western platforms, China's Alipay and WeChat Pay, both structured around QR code based account to account and wallet mechanisms rather than traditional card rails, dominate domestic Chinese retail payments and have progressively expanded cross border acceptance, particularly for inbound tourism and cross border e-commerce, introducing a QR based, largely card free competitive model into markets that remain otherwise dominated by international card schemes.

4.4 Comparative Tables

Table 2: Summary of Major Global Payment Systems

System	Type	Established	Core Mechanism	Cross Border Cost
SWIFT	Messaging network	1973	Secure interbank instruction messaging; settlement via correspondent banking	Low messaging fee; settlement cost varies by corridor
Visa / Mastercard	Card network	1958 / 1966	Four party card authorisation, clearing, and settlement	Approximately 1% network fee plus 1% to 3% merchant fee
PayPal / Stripe	Digital wallet / payment gateway	1998 / 2010	Online consumer and merchant payments layered on card and bank rails	Approximately 2.9% to 4.4% plus a fixed fee, varying by market
Alipay / WeChat Pay	Super app wallet	2004 / 2013	QR code based account to account and wallet payments	Low to moderate, market dependent

Table 3: Technical Benchmark Comparison of Global Payment Systems

Characteristic	SWIFT / Correspondent Banking	Card Networks (Visa/Mastercard)	Digital Wallets (PayPal/Alipay)
Typical settlement latency	Hours to several business days per corridor	Authorisation instant; clearing and settlement in daily batches	Instant to consumer view; underlying settlement follows card or bank rail timing
Foreign exchange margin	Set by correspondent banks; historically opaque, improving under gpi	Card network conversion rate plus issuer markup, typically 1% to 3%	Platform set conversion margin, typically 2% to 4%

Characteristic	SWIFT / Correspondent Banking	Card Networks (Visa/Mastercard)	Digital Wallets (PayPal/Alipay)
Security protocol	Encrypted standardised messaging; sanctions and AML screening at each hop	EMV chip, tokenisation, 3-D Secure for card not present transactions	Platform level fraud scoring, tokenisation, two factor authentication
Throughput characteristic	Designed for high value, lower volume interbank messages	Designed for very high volume, low to moderate value retail transactions	High volume, low to moderate value consumer transactions

CHAPTER 5: LOCAL SECTOR ANALYSIS: CASE STUDY OF NEPAL AND REGIONAL SYSTEMS

Local payment systems are domestic infrastructures, typically established or promoted by a national central bank, that enable fast and low cost transfers within a country. Many of these systems have been developed specifically to advance financial inclusion, reduce dependence on cash, and lower the cost of everyday transactions for individuals and small merchants. This chapter presents detailed case studies of India's UPI and Brazil's Pix as internationally recognised reference models, before turning in depth to Nepal's own domestic payment infrastructure.

5.1 Case Study: UPI (India)

UPI was introduced in 2016 by the National Payments Corporation of India (NPCI), a not for profit entity promoted by the Reserve Bank of India and the Indian Banks' Association, and allows users to transfer funds instantly between bank accounts using a virtual payment address (VPA), a mobile number, or a QR code, without needing traditional card infrastructure. Architecturally, UPI functions as a switch that sits above participating banks' core banking systems, routing a payment instruction from the payer's bank to the payee's bank in real time and confirming settlement within seconds, with NPCI itself acting as the payment system operator responsible for the underlying switch and settlement guarantee.

A defining economic feature of UPI is its zero merchant discount rate (MDR) policy on person to merchant transactions, mandated by the Indian government to accelerate merchant adoption; because merchants pay no per transaction fee to accept UPI payments, adoption among small merchants who would otherwise find card acceptance costly has been extremely rapid, and UPI now accounts for the large majority of India's electronic retail transaction volume by count. India has also developed a domestic card network, RuPay, which operates alongside UPI under NPCI and charges considerably lower merchant fees than the international card networks, further reducing India's structural reliance on foreign card schemes for domestic transactions.

5.2 Case Study: Pix (Brazil)

Pix was launched in November 2020 by the Central Bank of Brazil (Banco Central do Brasil) as a mandatory instant payment system, with participation required of all sufficiently large financial institutions operating in Brazil, enabling free or low cost transfers between individuals and businesses at any time of day, seven days a week. Technically, Pix relies on a centrally

operated directory (the DICT) that maps simple identifiers, a phone number, an email address, a tax identification number, or a randomly generated key, to the underlying bank account, allowing a payer to initiate a transfer without knowing the recipient's full account and routing details.

Within a short period after launch, Pix transaction volumes surpassed those of conventional payment methods such as debit cards and boletos (a traditional Brazilian payment slip) in the country, and the system is widely credited with accelerating financial inclusion among previously underbanked segments of the population by removing the need for a credit history or card issuance to transact digitally. The mandatory participation requirement imposed by the Central Bank, rather than a purely voluntary or market driven rollout, is frequently cited as a key structural factor explaining Pix's unusually rapid path to near universal domestic acceptance compared with voluntarily adopted instant payment systems in other jurisdictions.

5.3 Case Study: Nepal's Domestic Payment Infrastructure

Role of Nepal Rastra Bank and the Payment and Settlement Act

Nepal Rastra Bank (NRB), the country's central bank, is the primary regulatory authority for the payments sector under the Payment and Settlement Act, 2019, and the associated Payment and Settlement Bylaw, 2020. This legal framework establishes the two tier licensing structure described in Chapter 2, distinguishing payment system operators, licensed to run the underlying clearing, settlement, and switching infrastructure, from payment service providers, licensed to offer payment services such as digital wallets and merchant acquiring on top of that infrastructure, and it grants NRB supervisory authority over both categories, including licensing, capital adequacy, and consumer protection requirements.

Nepal Clearing House Limited: NCHL-IPS, connectIPS, and the National Payments Interface

Nepal's domestic payment ecosystem has developed around Nepal Clearing House Limited (NCHL), a company promoted by Nepal Rastra Bank together with commercial banks, development banks, finance companies, and Smart Choice Technologies. NCHL operates the Interbank Payment System (NCHL-IPS), launched in 2016, which facilitates domestic fund transfers between accounts held at different banks and financial institutions, supporting both direct debit and direct credit transaction types on a real time basis.

In 2018, NCHL launched connectIPS, a standardised platform that allows customers to link their bank accounts directly and make online fund transfers, utility bill payments, government revenue payments, credit card bill payments, and mobile wallet top ups without needing to hold a balance in an intermediary wallet. connectIPS is supported by dozens of banks and financial institutions and has since been extended with the National Payments Interface (NPI), a consolidated set of application programming interfaces that allows mobile wallets, insurance companies, remittance companies, the stock exchange, and government entities to connect to the underlying payment rails through a common, standardised integration layer rather than negotiating bespoke connections with each individual bank.

Digital Wallets and Payment Service Providers

Alongside connectIPS, licensed payment service providers such as eSewa, established in 2009 as Nepal's first licensed payment service provider, and Khalti, launched in 2017 and later merged with IME Pay, have built large user and merchant networks offering wallet based payments, QR transactions, and bill payment services. Both providers maintain extensive agent networks, physical locations, often small shops or kiosks, where customers without smartphones or bank accounts can deposit or withdraw cash against their wallet balance, a distribution model that has been important in extending digital payment access into Nepal's less urbanised areas.

Interoperability: NEPALPAY QR and Unified Merchant Acquiring

Interoperability across Nepal's various banks and payment service providers has been substantially advanced by NEPALPAY QR, a shared QR code standard introduced under NRB's coordination that allows customers of different banks and payment service providers to pay any participating merchant using a single QR code, rather than requiring merchants to display separate, incompatible QR codes for each provider. This unified acquiring model, in which a merchant maintains one QR code capable of accepting payments regardless of which bank or wallet the customer holds, mirrors the interoperability principle underlying UPI in India and is considered central to Nepal's ability to extend digital payment acceptance to small and micro merchants without imposing multiple, redundant onboarding processes.

5.4 Regional and Global Alternatives

Beyond the systems examined above, several other domestic and regional instant payment systems provide useful comparative reference points. SEPA Instant Credit Transfer is a euro

area initiative enabling instant transfers across participating European countries within a harmonised regulatory framework, illustrating that instant payment interoperability can be extended across, and not only within, national borders when countries share a common currency and a coordinating regulatory body. FedNow, introduced by the United States Federal Reserve in 2023, provides round the clock interbank instant payment infrastructure directly operated by the central bank, a model structurally similar in intent to Pix though introduced into a market with a longer established, more fragmented private sector payment landscape. M-Pesa, launched in Kenya in 2007 by Safaricom, remains one of the most influential mobile money platforms globally, demonstrating that basic mobile phone based person to person transfer, without a smartphone, a bank account, or even internet data connectivity, can achieve transformative financial inclusion outcomes in markets with limited traditional banking infrastructure, a lesson of continuing relevance to Nepal's own agent network based inclusion strategy.

5.5 Structured Tables

Table 4: Summary of Major Local and Regional Payment Systems

System	Country / Region	Launch Year	Operator	Core Mechanism
UPI	India	2016	National Payments Corporation of India	Account to account, mobile number / QR
Pix	Brazil	2020	Central Bank of Brazil	Account to account, instant, 24/7, mandatory participation
connectIPS / NCHL-IPS	Nepal	2016 / 2018	Nepal Clearing House Limited	Direct bank account linkage, real time
SEPA Instant	European Union	2017	European Payments Council / national banks	Account to account, euro area
FedNow	United States	2023	U.S. Federal Reserve	Account to account, interbank, 24/7
M-Pesa	Kenya (and other markets)	2007	Safaricom / Vodafone	Mobile money, SIM based wallet

Table 5: Licensed Payment Service Providers and Operators in Nepal

Provider	Role	Year	Tech Stack	Key Feature
Nepal Clearing House Limited	Payment System Operator	2016	NCHL-IPS, connectIPS, National Payments Interface	Direct bank account linkage; government revenue collection

Provider	Role	Year	Tech Stack	Key Feature
eSewa	Payment Service Provider (wallet)	2009	Proprietary wallet platform with agent network integration	First licensed digital wallet in Nepal; broad agent and merchant network
Khalti	Payment Service Provider (wallet)	2017	Wallet platform merged with IME Pay; merchant API	QR payments and e commerce API integration
NEPALPAY QR	Interoperable QR standard	2021	NRB coordinated shared QR specification	Single QR code accepted across participating banks and PSPs

CHAPTER 6: COMPARATIVE ANALYSIS

Global and local payment systems differ across several dimensions, though both continue to serve important and largely complementary roles within the broader payments landscape. This chapter presents a structured comparison across five criteria, extending the case studies of Chapters 4 and 5 into a consolidated analytical framework.

6.1 Total Cost of Transaction and Interchange (Global versus Local)

Global card networks and cross border transfer mechanisms generally carry higher fees, reflecting the cost of currency conversion, cross border settlement, correspondent banking intermediation, and the interchange economics of the four party card model. Local instant payment systems, being domestically operated and, in the cases of UPI, Pix, and connectIPS, promoted or mandated by central banks or public private utilities with an explicit financial inclusion objective, tend to offer substantially lower costs, and in several cases, notably UPI's zero MDR policy and Pix's free consumer transfers, are free of charge for individual users and small merchants.

6.2 Settlement Speed, Availability, and Latency

Local real time payment systems typically settle transactions within seconds and operate continuously, including weekends and public holidays, because they are architecturally designed around a single domestic settlement guarantee rather than a chain of independent correspondent relationships. Cross border payments through global networks, while improving through initiatives such as SWIFT gpi and ISO 20022 migration, can still take from a few hours to several business days, depending on the countries, currencies, and number of intermediary banks involved in the transaction chain.

6.3 Accessibility and Financial Inclusion Impact

Local payment systems linked to mobile phone numbers or QR codes, such as UPI, Pix, connectIPS, and Nepal's wallet ecosystem, have proven effective in extending financial services to populations with limited access to traditional banking or credit cards, since a QR code or a mobile number is a far lower barrier to entry than card issuance, which itself typically presupposes an existing bank account and, for credit cards, a credit assessment. Global card networks, by contrast, require underlying card issuance and, in many cases, credit assessment infrastructure that can limit accessibility in lower income segments of the population, a

structural factor that has shaped the divergent inclusion trajectories of card dominant and QR or A2A dominant markets.

6.4 Regulatory Oversight and Cross Border Sovereignty

Local payment systems are generally subject to a single national regulator, such as Nepal Rastra Bank, the Reserve Bank of India, or the Central Bank of Brazil, which allows for rapid policy adjustment, direct alignment with domestic financial inclusion goals, and, in Pix's case, the ability to mandate participation across the banking sector. Global payment systems must comply with multiple, sometimes divergent, regulatory regimes across the jurisdictions in which they operate, including anti money laundering, sanctions screening, and data localisation requirements, which increases compliance complexity but also provides broader oversight consistency for genuinely cross border transactions, and introduces geopolitical considerations, such as sanctions driven access restrictions, that domestic systems are largely insulated from.

6.5 Master Comparative Analysis Matrix

Table 6: Master Comparative Analysis Matrix

Criteria	Global Payment Systems	Local Payment Systems
Geographic reach	Cross border, multi currency	Domestic, single currency
Typical settlement time	Minutes to several business days	Seconds to a few minutes
Typical cost to end user	Moderate to high (FX and network fees)	Low to none
Regulatory oversight	Multiple national regulators and standard setters	Single national regulator, usually the central bank
Financial inclusion impact	Indirect; generally requires card or bank access	Direct; often mobile number or QR based
Availability	Business hours in some corridors; improving under gpi	Typically twenty four hours, seven days a week
Primary use case	International trade, travel, remittances	Domestic retail, bill payment, peer to peer transfer

6.6 Illustrative Transaction Scenarios

Table 7: Illustrative Transaction Scenarios

Scenario	Typical Rail	Approximate Settlement Time	Approximate Cost Structure
Cross border B2B invoice payment (Nepal to a European supplier)	SWIFT messaging with correspondent bank settlement	One to five business days	Correspondent bank fees, FX margin, possible intermediary deductions

Scenario	Typical Rail	Approximate Settlement Time	Approximate Cost Structure
Domestic QR merchant checkout (Kathmandu retail purchase)	NEPALPAY QR via connectIPS or a licensed wallet	Near instant (seconds)	Negligible or no fee to the consumer; low merchant fee where applicable
International remittance to a Nepali bank account	SWIFT or dedicated remittance operator, terminating in NCHL-IPS	Minutes to two business days depending on corridor	Sending fee plus FX margin; domestic leg is low cost
Person to person transfer between two Nepali bank customers	NCHL-IPS / connectIPS	Near instant (seconds)	Low or no fee

CHAPTER 7: RISKS, SECURITY, AND OPERATIONAL CHALLENGES

7.1 Instant Payment Fraud, Social Engineering, and Irreversibility

A structural risk shared across instant payment systems, including UPI, Pix, and NCHL-IPS, is that the same speed and finality that make them attractive also make fraudulent transfers effectively irreversible once completed, since funds can be withdrawn or moved onward by the receiving party within seconds of receipt, leaving little or no window for a victim or their bank to intervene before value has left the system. This has made social engineering, in which a fraudster persuades a victim to authorise a transfer voluntarily under false pretences, a leading category of loss on instant payment rails internationally, in contrast to card fraud, which more often involves unauthorised use of stolen credentials that a cardholder did not knowingly approve.

7.2 Cybersecurity, System Outages, and Single Point of Failure or Concentration Risk

Because instant payment systems and card networks both rely on a small number of centrally operated switches, NPCI for UPI, the Central Bank of Brazil's infrastructure for Pix, and NCHL for connectIPS and NCHL-IPS in Nepal, an outage or a successful cyberattack against that central switch can disrupt a very large share of a country's electronic retail payments simultaneously, a pattern of concentration risk that does not arise in the same way for less centralised payment methods such as cash or, to a lesser extent, individually issued cards. This concentration risk must be weighed against the substantial efficiency and interoperability gains that a single shared switch provides relative to a fragmented set of bilateral bank connections.

7.3 Compliance Overhead: AML/CFT, KYC, Sanctions Screening, and Geopolitical Risk

Global payment systems carry substantial compliance overhead arising from anti money laundering and combating the financing of terrorism (AML/CFT) obligations, know your customer (KYC) requirements, and sanctions screening obligations that must be applied consistently across every jurisdiction a transaction passes through, a burden that grows with the number of intermediary institutions in a correspondent banking chain. Geopolitical risk is a distinctive feature of global systems: access to SWIFT messaging or to major card networks can be, and in specific documented cases has been, restricted as an instrument of international

sanctions policy, a form of systemic risk that domestic payment systems, operating entirely within a single sovereign jurisdiction, are largely insulated from, though domestic systems remain exposed to their own forms of political and regulatory risk within that jurisdiction.

7.4 Risk and Mitigation Matrix

Table 8: Risk and Mitigation Matrix

Risk Category	Relevant To	Typical Mitigation
Social engineering and authorised push payment fraud	Both, acute for instant A2A rails	Real time transaction monitoring, payee name verification, cooling off periods for new payees, user education
Settlement and counterparty risk	Global (correspondent banking)	Correspondent due diligence, pre funding (nostro) arrangements, SWIFT gpi tracking
System outage or concentration risk	Local (centralised switches)	Redundant infrastructure, multiple participating institutions, business continuity planning
Regulatory and compliance risk (AML/CFT, sanctions)	Both, more complex for global	Ongoing adherence to central bank directives and international standards, automated screening
Currency and exchange rate risk	Global	Hedging arrangements, transparent FX pricing under initiatives such as gpi
Plugin/API or third party integration vulnerabilities	Local (PSP and merchant integrations)	Security audits of merchant and wallet integrations, API key rotation, penetration testing

CHAPTER 8: EMERGING TRENDS AND FUTURE OUTLOOK

Several trends are likely to shape the future relationship between global and local payment systems. Central banks in multiple regions are examining bilateral and multilateral linkages between domestic instant payment systems, exemplified by early UPI linkages between India and Singapore and between India and the United Arab Emirates, to enable faster and cheaper cross border transfers without relying solely on traditional card networks or correspondent banking. Nepal's own NCHL infrastructure, having already achieved substantial domestic interoperability through connectIPS and NEPALPAY QR, represents a plausible future candidate for similar bilateral fast payment linkages with major remittance corridor partners, given the importance of remittance inflows to Nepal's economy.

Central bank digital currencies (CBDCs) are being actively explored by a growing number of monetary authorities as a potential complement to, rather than a wholesale replacement for, existing instant payment infrastructure, with particular attention to offline digital payment capability that would allow transactions to be completed without an active internet or mobile data connection, a feature of specific relevance to markets, including parts of Nepal, where network connectivity is not uniformly reliable. The industry wide migration to the ISO 20022 messaging standard, discussed in Chapter 4, is expected to continue over the coming years across SWIFT and numerous domestic RTGS systems, progressively improving straight through processing rates and enabling richer, more structured remittance data to travel with each cross border payment.

Artificial intelligence driven real time fraud mitigation is being integrated increasingly deeply into both global card network risk engines and domestic instant payment switches, using behavioural and transaction pattern analysis to flag suspicious transfers before completion rather than relying solely on post transaction dispute processes, a development of particular importance for instant payment systems given the irreversibility risk discussed in Chapter 7. For Nepal specifically, continued expansion of NEPALPAY QR acceptance, deeper integration between connectIPS and international remittance operators, and gradual regulatory attention to CBDC feasibility by Nepal Rastra Bank are the most immediately relevant developments to monitor over the medium term.

CHAPTER 9: CONCLUSION AND STRATEGIC SUMMARY

Global and local payment systems serve distinct but complementary functions within the financial ecosystem. Global systems such as SWIFT, Visa, and Mastercard provide the standardised messaging, currency conversion, and multi jurisdictional compliance infrastructure necessary for cross border trade, travel, and remittances, functions that no purely domestic system can presently replicate on its own. Local systems such as UPI, Pix, and Nepal's connectIPS and wallet ecosystem have, in turn, demonstrated the ability to deliver fast, low cost, and financially inclusive domestic payment services precisely because they are optimised for a single currency, a single regulatory framework, and, in several cases, a central bank mandate or guarantee that a purely commercial global network is not structured to provide.

Rather than one category displacing the other, the evidence assembled in this report suggests that the payments industry, both globally and within Nepal, is moving toward greater interoperability, in which local systems handle the substantial majority of domestic retail transactions while modernised global networks continue to underpin cross border commerce, increasingly connected through bilateral fast payment linkages, a shared ISO 20022 messaging language, and, potentially, central bank digital currency infrastructure. For Nepal, the strategic balance to be struck is between continued domestic autonomy and technical sovereignty over its clearing and settlement infrastructure, exercised through Nepal Rastra Bank and NCHL, and deeper, more efficient integration with the global rails on which the country's substantial remittance inflows and cross border commerce continue to depend. Countries and institutions that succeed in linking these two layers efficiently, without compromising domestic regulatory control or systemic resilience, are likely to see the greatest gains in both domestic financial inclusion and international trade competitiveness over the coming decade.

CHAPTER 10: ACTIONABLE RECOMMENDATIONS

Recommendations arising from this study are grouped below by stakeholder category, reflecting the differing mandates, resources, and risk exposure of each group.

10.1 For Central Banks and Regulators (Nepal Rastra Bank and International Equivalents)

- Continue to strengthen the two tier payment system operator and payment service provider licensing framework established under the Payment and Settlement Act, ensuring proportionate but consistent oversight across all licensed entities.
- Evaluate participation in bilateral or regional fast payment linkage initiatives, building on the interoperability already achieved domestically through connectIPS and NEPALPAY QR, to reduce dependence on costlier correspondent banking channels for remittance corridors.
- Advance feasibility assessment of a central bank digital currency, with particular attention to offline payment functionality suited to areas with limited network connectivity.
- Strengthen mandated fraud monitoring, payee verification, and consumer protection standards for instant payment service providers, given the irreversibility risk associated with real time transfers.

10.2 For Domestic Payment Operators (NCHL, PSPs, and PSOs)

- Continue expanding NEPALPAY QR acceptance among small and micro merchants to extend the financial inclusion benefits already realised among larger urban merchants.
- Invest in redundant, geographically distributed infrastructure to mitigate the concentration risk inherent in a small number of centrally operated switches.
- Pursue direct technical integration with major international remittance operators to shorten the settlement path between an incoming cross border remittance and its final domestic credit through connectIPS or NCHL-IPS.
- Adopt or prepare for ISO 20022 message compatibility to align with the trajectory of SWIFT and international RTGS modernisation.

10.3 For Commercial Banks and Financial Institutions

- Maintain active participation in both connectIPS and correspondent banking relationships to serve customers across both domestic and cross border payment needs.
- Invest in real time transaction monitoring systems capable of detecting social engineering and authorised push payment fraud patterns specific to instant transfer rails.
- Provide clear, proactive customer communication regarding the irreversibility of instant domestic transfers, distinguishing this risk profile from the dispute and chargeback protections more commonly associated with card transactions.

10.4 For Merchants and End Users

- Adopt NEPALPAY QR or an equivalent interoperable QR standard to accept payment from customers of any participating bank or wallet without maintaining multiple, incompatible QR codes.
- Exercise heightened caution before authorising instant transfers to previously unknown payees, given the practical difficulty of reversing a completed instant payment.
- For businesses engaged in cross border trade, compare the total landed cost, including network fees, merchant fees, and foreign exchange margins, of card, SWIFT, and digital wallet alternatives before selecting a payment channel for a given transaction corridor.

Table 9: Stakeholder Priority Action Table

Stakeholder Category	Top Priority Recommendation
Central Banks and Regulators	Strengthen licensing oversight and evaluate cross border fast payment linkages while advancing CBDC feasibility work.
Domestic Payment Operators (NCHL, PSPs, PSOs)	Expand NEPALPAY QR acceptance among micro merchants and invest in redundant switch infrastructure.
Commercial Banks and Financial Institutions	Deploy real time fraud monitoring calibrated to the irreversibility risk of instant transfers.
Merchants and End Users	Adopt interoperable QR acceptance and exercise caution when authorising transfers to new payees.

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